

Lab 05: Black Box Test Plan

Test Overview

To run the tests:

1. Right click on PackSchedulerGUI class in the Package Explorer.
2. Select Run As > Java Application

All tests files should be located in the test-files/ folder.

The project should have a registrar.properties file that contain values for the following properties:

first=
last=
id=
email=
pw=

The values should be substituted in the tests below as appropriate for the local copy of the registrar.properties file.

If you run all tests independently, you can close the GUI between each test. However, the tests are structured so that they may be run in order without shutting down the GUI.

Test ID	Description	Expected Results	Actual Results
T1: Invalid Login	Preconditions: PackSchedulerGUI has loaded User ID: Iberg Password: pw Click Login	Pop up with message "User	Pop up with message "User doesn't exist." User ID and password text

	Click OK Click Clear	doesn't exist." User ID and password text fields are cleared.	fields are cleared.
T2: Registrar Login	Preconditions: PackSchedulerGUI has loaded User ID: registrar.properties.id Password: registrar.properties.pw Click Login	View switches to Registrar panel with Student Directory functionality displayed	View switches to Registrar panel with Student Directory functionality displayed
T3: Load Student Directory	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2) Click Load Student Directory Select test-files/student_records.txt Click Select	10 students are listed in alphabetical order by last name. The list starts with Demetrius Austin and ends with Griffith Stone.	
T4: Add Invalid Student - no first name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu		Error Message: Invalid first name

	• Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid first name	
T5: Add Invalid Student - no last name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid last name	
T6: Add Invalid Student - no id	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw	Error Message: Invalid id	

	Max Credits: 15 Click Add Student Click OK	Error Message: Invalid id	
T7: Add Invalid Student - no email	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: - First Name: Patience - Last Name: Wilkins - ID: pwilkins - Email: - Password: pw - Repeat Password: pw - Max Credits: 15 Click Add Student Click OK	Error Message: Invalid email	Error Message: Invalid email
T8: Add Invalid Student - invalid email	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: - First Name: Patience - Last Name: Wilkins - ID: pwilkins - Email: pwilkinsncsu.edu - Password: pw - Repeat Password: pw - Max Credits: 15 Click Add Student	Error Message: Invalid email	Error Message: Invalid email

	Click OK		
T9: Add Invalid Student - password s don't match	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw1 • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Passwords do not match	Error Message: Passwords do not match
T10: Add Invalid Student - zero max credits	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 0 Click Add Student Click OK	Student added with max credits of 18.	Not sure

T11: Add Invalid Student - 11 max credits	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 19 Click Add Student Click OK	Student added with max credits of 18	Not sure
T12: Add Student - end	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Patience Wilkins is added to the Student Directory and is the last student in the list.	Patience Wilkins is added to the Student Directory and is the last student in the list.
T13: Add Student -	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run.		

middle	Enter the following in the Student Information section: • First Name: Rylee • Last Name: Puckett • ID: rpuckett • Email: rpuckett@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Rylee Puckett is added to the Student Directory between Dylan Nolan and Cassandra Schwartz	
T14: Add Student - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Melvin • Last Name: Acevedo • ID: macevedo • Email: macevedo@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Melvin Acevedo is added to the Student Directory and is the first student in the list.	
T15: Remove Student -	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2).		Error Message: No student selected

no selection	Click Remove Student	Error Message: No student selected	
T16: Remove Student - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Select Shannon Hansen Click Remove student	Shannon Hansen is removed. Athea Hicks follows Emerald Frost in the list	Shannon Hansen is removed. Athea Hicks follows Emerald Frost in the list
T17: Remove Student - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T14 has run. Select Melvin Acevedo Click Remove student	Melvin Acevedo is removed. Demetrius Austin is front of the directory	Melvin Acevedo is removed. Demetrius Austin is front of the directory
T18: Remove Student - back	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12 has run. Select Patience Wilkins Click Remove student	Patience Wilkins is removed. Griffith Stone is last in the directory	Patience Wilkins is removed. Griffith Stone is last in the directory
T19: Save Student	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12-T18 have run.	Contents should be (note that the password hash may be different due to encoding):	The file shows:

Directory	Click Save Student Directory Select test-files directory and name the file t19_student_directory.txt.	Demetrius,Austin,daustin,Cura bitur.egestas.nunc@placerator cilacus.co.uk,0ÉRú±"ÃùuŸ!Ù\7 X²F'pâ9•{-OîFâapÃ,18 Lane,Berg,lberg,sociis@non.or g,0ÉRú±"ÃùuŸ!Ù\7X²F'pâ9•{- OîFâapÃ,14 Raymond,Brennan,rbrennan,lit ora.torquent@pellentesquema ssalobortis.ca,0ÉRú±"ÃùuŸ!Ù\ 7X²F'pâ9•{-OîFâapÃ,12 Emerald,Frost,efrost,adipiscin g@acipsumPhasellus.edu,0É Rú±"ÃùuŸ!Ù\7X²F'pâ9•{-OîFâ apÃ,3 Althea,Hicks,ahicks,Phasellus. dapibus@luctusfelis.com,0ÉR ú±"ÃùuŸ!Ù\7X²F'pâ9•{-OîFâap Ã,11 Zahir,King,zking,orci.Donec@a metmassaQuisque.com,0ÉRú ±"ÃùuŸ!Ù\7X²F'pâ9•{-OîFâap Ã,15 Dylan,Nolan,dnolan,placerat.C ras.dictum@dictum.net,0ÉRú± "ÃùuŸ!Ù\7X²F'pâ9•{-OîFâapÃ, 5 Rylee,Puckett,rpuckett,rpucket t@ncsu.edu,0ÉRú±"ÃùuŸ!Ù\7 X²F'pâ9•{-OîFâapÃ,15 Cassandra,Schwartz,cschwartz, semper@imperdietornare.co.	
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T20: New Student Directory	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12-T18 have run. Click New Student Directory	An empty student directory is shown.	An empty student directory is shown.
T21: Add Student - first	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T20 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Patience Wilkins is added to the Student Directory and is the only student in the list.	Patience Wilkins is added to the Student Directory and is the only student in the list.
T22: Course Catalog view	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Click Course Catalog	Course Catalog functionality displayed	Course Catalog functionality displayed

T23: Load Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2) Click Load Course Catalog Select test-files/course_records.txt Click Select	13 courses are listed starting with CSC 116-001 and ending with CSC 316-001 as provided in the course_records.txt file (minus the one duplicate course)	13 courses are listed starting with CSC 116-001 and ending with CSC 316-001 as provided in the course_records.txt file (minus the one duplicate course)
T24: Add Invalid Course - no name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid course name	Error Message: Invalid course name
T25: Add Invalid Course - no title	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information		Error Message: Invalid course title

	section: • Course Name: CSC 236 • Course Title: • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid course title	
T26: Add Invalid Course - no section	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section number	Error Message: Invalid section number
T27: Add Invalid Course -	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run.		Error Message: Invalid section number

section too short	Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 0011 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section number	
T28: Add Invalid Course - section too long	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 01 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section number	Error Message: Invalid section number
T29: Add	Preconditions: PackSchedulerGUI has loaded and		

Invalid Course - no instructor id	registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid instructor unity id	Error Message: Invalid instructor unity id
T30: Add Invalid Course - end time after start time	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 11:30 AM • End Time: 10:15 AM • Meeting Days: Tuesday & Thursday Click Add Course	Error Message: Invalid course times	Error Message: Invalid course times

	Click OK		
T31: Add Invalid Course - no meeting days	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Click Add Course Click OK	Error Message: Invalid meeting days	Error Message: Invalid meeting days
T32: Add Course - end	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC 236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday		CSC 236-001 is added to the list between CSC 230-001 and CSC 316-001

	• End Time: 12:30 PM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	CSC 116-006 is added to the list between CSC 116-003 and CSC 216-001.	
T35: Remove Course - no selection	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Click Remove Course	Error Message: No course selected	
T36: Remove Course - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Select CSC 216-001 Click Remove Course	CSC 216-001 is removed. CSC 116-006 is followed by CSC 216-002.	CSC 216-001 is removed. CSC 116-006 is followed by CSC 216-002.
T37: Remove Course - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T33 has run. Select CSC 113-001 Click Remove Course	CSC 113-001 is removed. CSC 116-001 is the first course in the catalog	CSC 113-001 is removed. CSC 116-001 is the first course in the catalog
T38: Remove Course - back	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T32 has run. Select CSC 236-001		CSC 236-001 is removed. CSC 316-001 is the last course in the catalog

	Click Remove Course	CSC 236-001 is removed. CSC 316-001 is the last course in the catalog	
T39: Save Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T32-T38 have run. Click Save Course Catalog Select test-files directory and name the file t39_course_catalog.txt.	Contents should contain the courses in the following order with all information as in the original file or entered via earlier tests: CSC 116-001 CSC 116-002 CSC 116-003 CSC 116-006 CSC 216-002 CSC 216-601 CSC 217-202 CSC 217-211 CSC 217-223 CSC 217-601 CSC 226-001 CSC 230-001 CSC 316-001	CSC 116-001 CSC 116-002 CSC 116-003 CSC 116-006 CSC 216-002 CSC 216-601 CSC 217-202 CSC 217-211 CSC 217-223 CSC 217-601 CSC 226-001 CSC 230-001 CSC 316-001
T40: New Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12-T18 have run. Click New Course Catalog	An empty course catalog is shown.	An empty course catalog is shown.
T41: Add Course - first	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T20 has run.		CSC 216-601 is added to the list

	Enter the following in the Course Information section: • Course Name: CSC 216 • Course Title: Programming Concepts - Java • Course Section: 601 • Credits: 4 • Instructor Id: jep • Start Time: 0 • End Time: 0 • Meeting Days: Arranged Click Add Course Click OK	CSC 216-601 is added to the list	
T42: Login Student	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T21 has run Click Logout User ID: pwilkins Password: pw Click: Login	User is returned to the Login view pwilkins is logged in and is switched to the user view. All that is shown is a Logout button.	User is returned to the Login view pwilkins is logged in and is switched to the user view. All that is shown is a Logout button.